UX / PRODUCT DESIGN CASE STUDY · 2026

Calmly: one app, two calmer worlds.

An accessible companion for people who are blind or have low vision, and for autistic people and the families who support them. Voice-first in one mode, low-stimulus in the other, and built for Nigerian streets, money and languages.

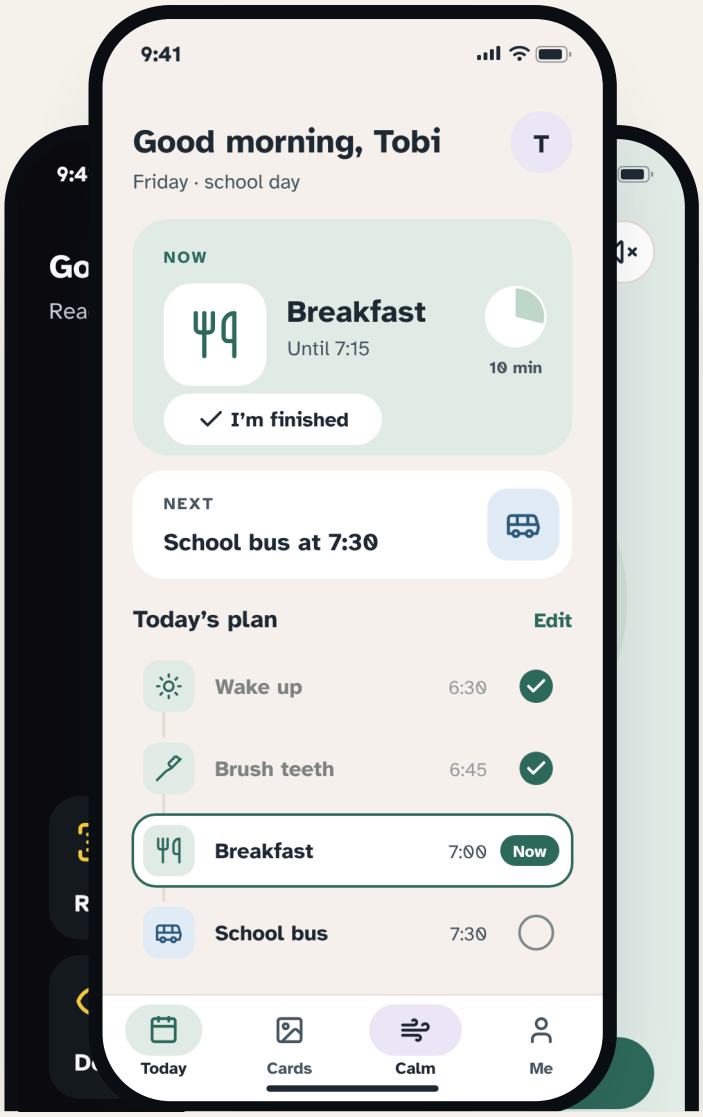
Designed by **Clement Olatunde**

ROLE
UX research, UX/UI design

PLATFORM
Mobile, Android and iOS

DELIVERABLES
10 hi-fi screens, flows, design system

STANDARD
WCAG 2.2 AA (AAA where it counts)



01

The problem

Assistive apps tend to come one disability at a time, priced in dollars, tuned for fast data and Western defaults. A blind trader in Lagos can't get an app to read a ₦500 note. An autistic child in Akure gets a bright, noisy kids' app when what he needs is to know what happens next.

How might we give blind, low-vision and autistic people one calm, dignified companion that works on the phone they already have, in the language they speak, with or without data?

02

Desk research

A quick desk review of public health data and the apps people already use. Every number names its source. Figures marked **UNVERIFIED** come from a search summary, not the full source page.

2.2 billion+

people worldwide have a near or distance vision impairment. At least 1 billion of those cases could have been prevented or have not been addressed.

WHO, Blindness and vision impairment fact sheet

4.2%

of Nigerian adults aged 40+ are blind, and 84% of that blindness is avoidable.

Abdull et al., Nigeria National Blindness and Visual Impairment Survey, IOVS 2009 (fieldwork 2005–07) **UNVERIFIED**

≈1 in 127

people worldwide were autistic in 2021.

WHO, Autism fact sheet **UNVERIFIED**

1 in 31

8-year-olds in the US have been identified as autistic. Nigeria has no national figure yet; local studies vary widely.

CDC, ADDM Network data **UNVERIFIED**; Bakare et al., ASD in Nigeria: a scoping review, 2019

What already exists, and the gap

APP	WHAT IT DOES WELL	GAP FOR OUR USERS
Be My Eyes	Free. Video calls to sighted volunteers (its site shows about 11 million volunteers and over 1 million blind users), plus AI descriptions.	Needs a live video connection, which is costly on mobile data. Nothing for routines or overload.

APP	WHAT IT DOES WELL	GAP FOR OUR USERS
Seeing AI (Microsoft)	Free. Reads text and scenes. Currency recognition launched with US, Canadian, UK and euro notes; Australian notes were added in 2024.	We found no source that lists Naira support UNVERIFIED. Not designed for autistic users.
Proloquo2Go	Gold-standard AAC symbol communication.	Listed at US$124.99 on the App Store during a 50%-off sale (Oct 2026), so about US$250 normally. iPhone and iPad only.
Tiimo	Visual planner and timers, built with ADHD and autism in mind.	Subscription. English-first. No picture-card communication or calm-down support.

Opportunity: one affordable, low-data companion that combines money and text reading, visual routines, picture cards and a calm button, with local currency, local voices and a caregiver mode.

03

Personas

Proto-personas built from the desk research. They're assumptions to test with real users, not interview findings.

E

Emeka, 34

Phone-accessories trader, Ikeja, Lagos

“I don’t want to hand my money to a stranger to count it.”

Glaucoma took most of his sight at 29. He uses TalkBack on a mid-range Android phone and buys data in small bundles.

NEEDS

- Count cash and read bank SMS alerts on his own
- Fast, hands-free actions while serving customers
- Works offline, with a voice that says “naira” properly

FRUSTRATIONS

- Apps with tiny, unlabeled buttons
- Video-call help that drains his data

T

Tobi, 9 & his mum Bisi

Primary-school pupil and teacher, Akure

“If he knows what’s coming, the whole morning goes better.” — Bisi

Tobi is autistic. When things get loud he struggles to talk, so pictures help. Bisi sets things up and wants to know when he needs her.

NEEDS

- See what’s now and what’s next, in pictures
- A way to say “I need a break” without words
- Alerts to a trusted adult when he’s overwhelmed

FRUSTRATIONS

- Surprise changes, noisy and flashy children’s apps
- AAC tools priced in dollars, on iPad only

04

Key user flows

Count money (vision mode)

1. “Hey Calmly, count money”
2. Camera finds the note on its own
3. Speaks “₦1,000” and gives 3 pulses
4. Keeps a running total
5. “Done” reads out the total

I’m overwhelmed (calm mode)

1. Tap Calm, always one tap away
2. “It’s okay.” Pick one option
3. Breathe for a minute with soft vibration
4. “I feel better” takes you back to your day
5. “Still need help” messages Mum

Morning routine

1. Today shows the Now card
2. Open the routine steps
3. A silent ring timer for each step
4. Heads-up 15 minutes before a change
5. “I did it” ticks the step off

Caregiver setup

1. “I’m setting this up for someone”
2. Pick the mode and preferences
3. Add trusted people
4. Build routines and picture cards
5. Lock settings with a PIN

05

Design principles

Say it, show it, buzz it

Every key event comes as voice, visuals and a haptic pattern, so nobody depends on one sense.

Big and forgiving

Touch targets are at least 48×48 dp, and main actions are 56–64 dp. That's well above WCAG 2.2's 24×24 px minimum (2.5.8 AA) and meets the 44×44 AAA target (2.5.5).

Contrast first

Text is at least 4.5:1 (1.4.3) and controls at least 3:1 (1.4.11). Vision mode goes past 7:1 (1.4.6 AAA). The lowest text pair in the system is 5.2:1.

No surprises

Reduce motion is on by default in calm mode, and the app follows the phone's own setting. No autoplay sound, no flashing (2.3.1). Changes get a heads-up first.

Screen-reader-first

Every control has a spoken name, role and state (4.1.2) and a logical focus order. The screens show TalkBack/VoiceOver captions as part of the design.

Low-data and dignified

Money and text are read on the phone itself. Voices download once. Caregiver features are always visible to the user, and location is never shared without asking.

06

Design system

Calm theme

Paper #F5F2EC
Background

Ink #1F2A33
Text · 13.1:1 on Paper

Ink 2 #4A5863
Secondary text · 6.6:1 on Paper

Sage #2F6B5E
Primary · white text 6.2:1

Sage tint #E3EEE8
Cards · Ink 12.3:1

Lavender #ECE8F5
Calm · Ink 12.1:1

Sky #E2ECF5
Routines · icon #2E5A7D 6.1:1

Sand #F4E9D8
Heads-up · icon #7A5B2E 5.2:1

Vision theme

Night #0A0D12
Background

White #FFFFFF
Text · 19.5:1 on Night

Mist #C9D1D9
Secondary · 12.6:1 on Night

Signal yellow #FFD23F
Actions, focus ring · 13.5:1

Surface #161B22
Tiles · yellow 12.0:1

Go green #5BE49B
Success · 10.7:1 on Surface

Type: Atkinson Hyperlegible

Designed by the Braille Institute for low-vision readers. Letters that look alike (I, l, 1; 0, O) are clearly different.

Display 36/44

Heading 28/36

Title 22/28

Body 17/24 (the minimum)

Caption 15/20

LABEL 13/16

Components

Voice button
180 dp, yellow on Night, static halo (no pulse)

Action tile
163×112, 36 dp icon, 19 pt label

Now / Next card
picture + time + quiet pie timer

Routine step
68 dp row, picture, number, check

Picture card
106×122, colour by category, speaks when tapped

Calm option
96 dp card, 60 dp picture tile

Buttons
56 dp (calm) / 60 dp (vision), pill shape

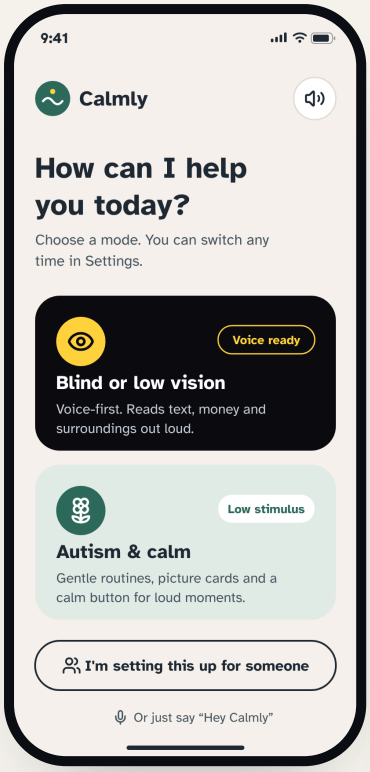
Toggle
52×32 with a 48 dp hit area

Tab bar
with a Calm button that's always there

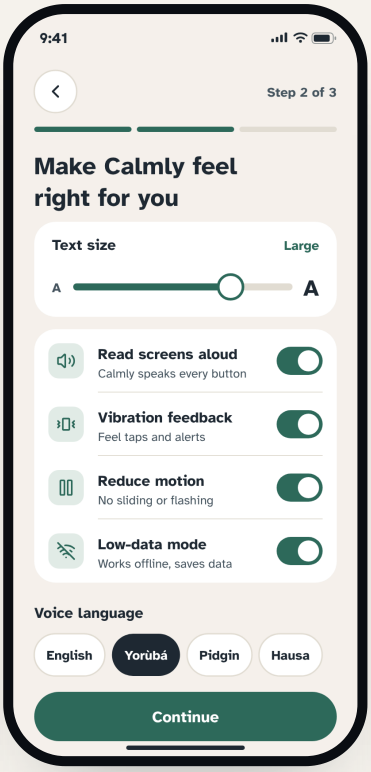
07

High-fidelity screens

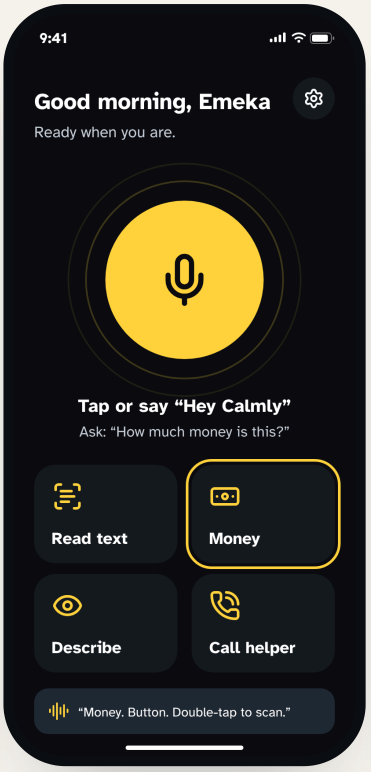
Ten screens at 390×844. Screens 03–04 are vision mode, 05–09 are calm mode.



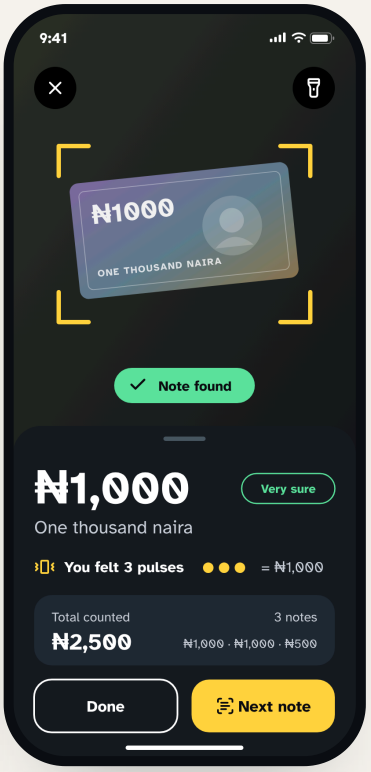
01 Welcome · choose a mode



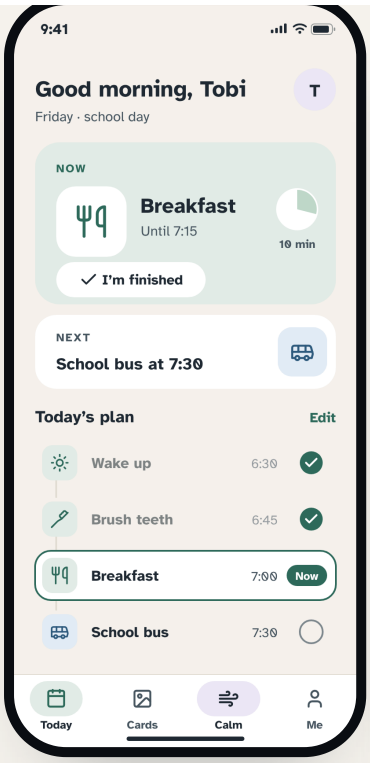
02 Onboarding · preferences



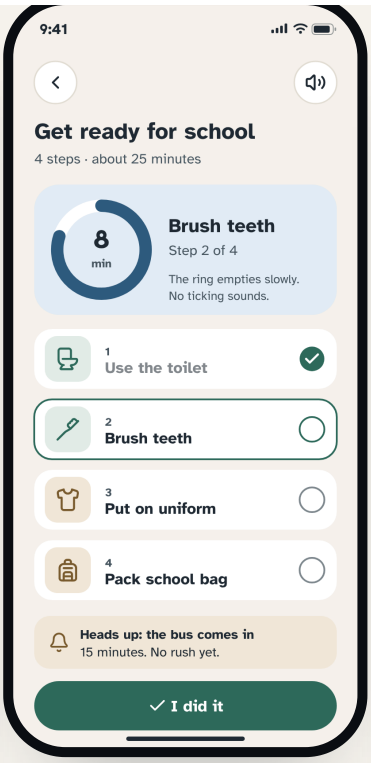
03 Vision · voice home



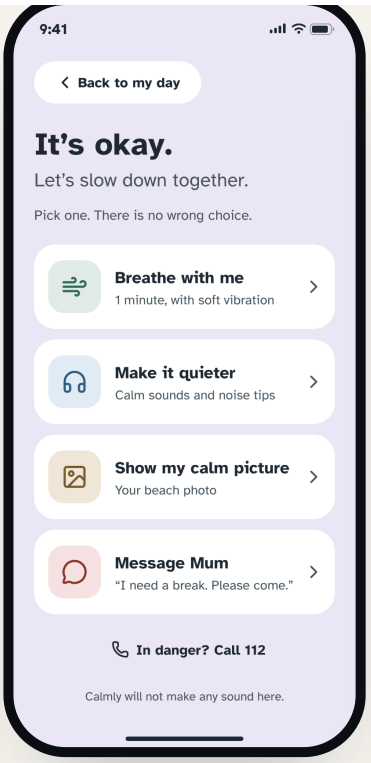
04 Vision · money identified



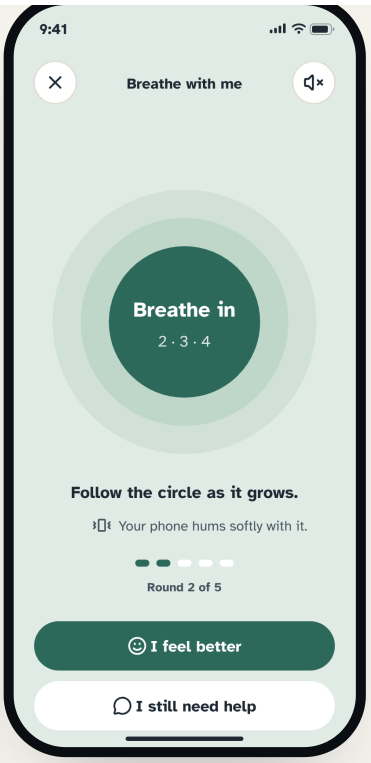
05 Calm · today



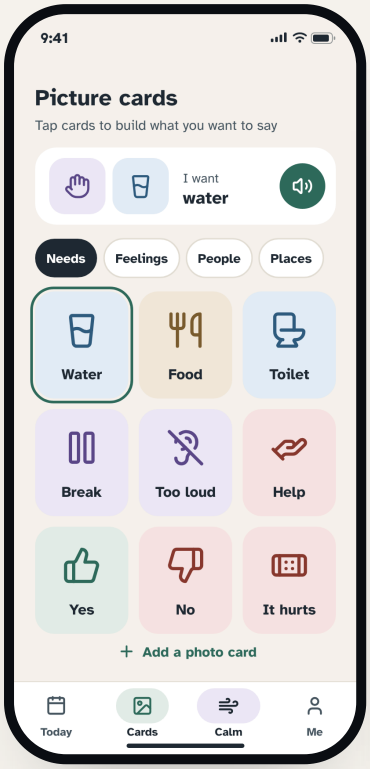
06 Calm · routine steps



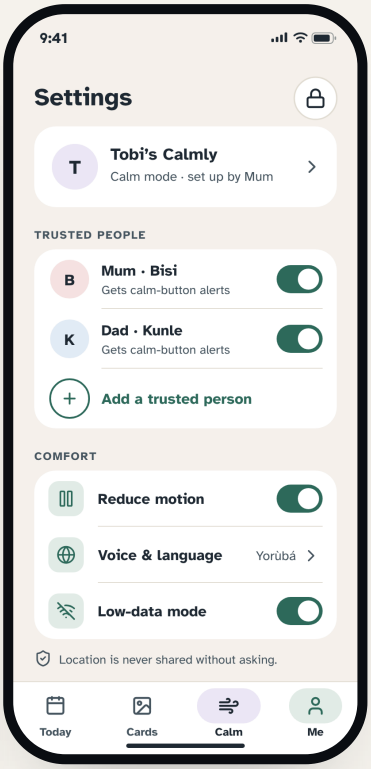
07 Calm button · options



08 Calm button · breathe



09 Calm · picture cards



10 Settings · caregiver

08

What I'd test next

- Usability sessions with blind and low-vision adults using TalkBack. Measure how often they count a set of notes correctly, and how long it takes.

- Co-design sessions with autistic young people, parents and special-needs teachers on picture cards and calm options.
- Recorded Yorùbá, Pidgin and Hausa voice prompts, checked by native speakers.
- Low-end Android performance and offline money recognition for every Naira note.

Designed by Clement Olatunde

UX/product designer · Google UX Design Certificate · Akure, Nigeria

Icons: Lucide (ISC licence). Type: Atkinson Hyperlegible (SIL OFL). Statistics belong to the sources linked above. This is a concept case study; personas are proto-personas.